

Single Source Justification (Simplified Acquisition)

Section A: General Contract Information

Contracting Activity: AFLCMC/PZIBA	Purchase Request/ Identification Number: F [REDACTED]
Project/Program Name: Thermal Vacuum Chamber and New Chiller Servicing and Installation	Estimated Contract Cost (including Options): [REDACTED]

Section B: Description of the Supplies/Services Required

1. The contractor shall service AFIT's custom thermal vacuum chamber with a solar simulator attachment (AFIT ABBESS TVAC Chamber: SN 51907) and liquid chiller (ABBESS SN: 51211) to restore capability to hold the required vacuum level (pressure <0.1mTorr) and temperature range (-49 to +66C) for flight hardware verification. In addition, the control system must be updated to use a more modern and supportable computer operating system (Windows 11). Contractor shall restore the equipment to proper operating condition
2. The system shall be integrated and tested to ensure leak free operation, both vacuum and refrigerant/coolant.
3. The system (chamber, solar simulator and chiller) shall be crated (using ABBESS provided crates) and shipped from AFIT to ABBESS and back to AFIT.

Section C: Justification for Soliciting from a Single Source

AFIT currently possesses a custom made thermal vacuum chamber (AFIT ABBESS TVAC Chamber) to simulate the space environment and chiller unit provided by ABBESS instruments to control its temperature. This contractor maintains knowledge and expertise on the equipment as the original manufacturer and the government has a responsibility to protect ABBESS's proprietary information in their chamber, chiller design, and control system/graphical user interface. The software application/user interface driving the control systems, and the hardware implementation of the controls for the chiller/heaters/vacuum pumps is ABBESS proprietary. ABBESS has the design documentation to allow revision of the systems, where other vendors would have to reverse engineer the systems--potentially interfering with ABBESS proprietary interests.

Of the seven (7) exceptions to full and open competition, it was determined that only one responsible source and no other supplies will satisfy the agencies need (FAR 6.302-1). The required thermal vacuum chamber is specialized equipment that is available from only one source, that is able to meet the specifications in the Statement of Work (SOW).

Section D: Efforts to Obtain Competition

Market Research was conducted, including outreach via phone and email to three different companies (PHPK, Thermal Product Solutions, and Kurt J. Lesker Co.) in addition to the original OEM (ABBESS). These companies are leading manufacturers and sellers of vacuum chambers, vacuum pumps, and related equipment, and the customer has previously purchased equipment from each of them. Of the three companies contacted, two responded (KJ Lesker and TPS), indicating that they were unwilling to refurbish or work on equipment they did not manufacture.

Additionally, a Sources Sought notice was posted to SAM.gov on 27 March 2025, with capabilities statements due by 2 April 2025. BHPE LLC expressed interest in the procurement; however, they failed to submit a capabilities statement by the due date. The OEM submitted their capabilities statement on 26 March 2025, and the customer determined that the OEM is capable of meeting the requirements outlined in the Statement of Work (SOW).

Based on this market research, it has been determined that ABBESS is the only vendor capable of fulfilling this requirement, justifying a sole-source award.

Section E: Steps to Preclude Future Single Source Awards

In future efforts to preclude future sole source awards, the Government will monitor the open market for vendors capable of fulfilling similar requirements and will post Sources Sought notices to maximize competition as much as possible.

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Section F: Contracting Officer's Determination	
I have determined, in accordance with FAR 13.106-1(b)(1) , that the circumstances of this contract action deem only a single source is reasonably available.	
Date	Typed Name and Rank/Grade of Contracting Officer
08-Apr-2025	